"Is Stephen Curry really a guard? A new perspective on player typologies using functional data analysis": Reviewer Responses

General/ Overall Responses

We thank the associate editor and the two reviewers for their thorough and detailed review of our manuscript. We have addressed each review point by point, making appropriate changes in the revised manuscript. As a result, we believe that the revised manuscript is greatly improved. Responses to each of the reviewer's points are contained in the remainder of this document. We look forward to your final decision with anticipation.

Associate Editor

Comment # 0.1

The author has effectively addressed some of the comments from the initial submission. Two new referees have reviewed the paper and give several suggestions for further revision. In addition, the authors should consider the following points.

Reply:

We thank the associate editor for the careful consideration of our manuscript. We have addressed their points and those of the two reviewers and improved the manuscript as a result. Our responses are presented in a point-by-point fashion below.

Comment # 0.2

The MFPCA approach was described more clearly in the revision, which leads to an important point about visualization. In equation (2), the eigenfunctions $\phi_k(s,t) \in \mathbb{R}^2$ are bivariate, with one component corresponding to made and the other to missed shots. Doing MFPCA instead of marginal FPCA is nice since variability in made and missed shot locations are taken together, yielding a single set of scores for each player to cluster. However, there is no indication in the figures regarding how much each eigenfunction loads onto made and missed shots. In other words, it is possible that $\phi_1(s,t)$ loads mostly onto made shots and almost none onto missed shots, or vice versa, but there is no way to tell this from the current analysis because the normalization (a linear transformation, I assume) is applied separately to every panel in each figure. It would be more informative to apply this transformation to the combined data across made and missed shots - that is, normalization is applied to each column. Also, it would be helpful to report how much each component contributes to the variability explained in made/missed shots separately, in addition to the joint figures currently reported.

Reply:

We agree with this comment. The separate normalization of each panel is useful for visualizing spatial patterns, but hides the relative magnitude of the made-shot and missed-shot components. We have therefore change the visualization. The normalization is now applied jointly to the made-shot and missed-shot surfaces within each principal components. More precisely, for each column of the figures, we computed the maximum value of the densities and divided the made-shot and missed-shot densities by this value. We do not see much difference with the previous version, indicating that the load of each eigenfunction onto made and missed shots is balanced.

Following Ramsay and Silverman, 2005, we reported $\|\phi_{kmissed}\|^2$ and $\|\phi_{k,made}\|^2$ to show how much each component contributes to the variability explained in made/missed shot. These quantities show the relative contribution in each dimension. This is a relative measure because

$$\|\phi_{k,\text{missed}}\|^2 + \|\phi_{k,\text{made}}\|^2 = 1,$$

by the orthonormality property.

COMMENT # 0.3

Although the author has argued that lack of consensus in a suitable spatial grid leads to subjectivity in previous analyses, there is no apparent justification for the choice of a 201×201, equally spaced grid for the MFPCA. In addition, the density estimation induces further subjectivity by choice of a bandwidth. Is there a way to assess sensitivity of the results and interpretations to these choices? It seems the choice is no less arbitrary for density representations compared to previous methods.

Reply:

We have added a sensitivity analysis to the revised manuscript. The 201×201 grid is used to evaluate the smooth kernel density estimates and approximate the MFPCA eigenfunctions. It does not define shooting regions, impose boundaries or create discrete shooting categories. To assess the numerical effect of this choice, we repeated the complete estimation procedure using two coarser grids, 51×51 and 101×101, and one finer grid, 401×401. The dominant spatial structures were stable across these resolutions. The mean functions and first eigenfunctions were very similar to those obtained with the 201×201 grid, and the 401×401 results were nearly indistinguishable from the main analysis. Some differences appeared for the coarsest 51×51 grid and for lower-variance components, which is expected because later components are more sensitive to numerical approximation. These results support the use of the 201×201 grid as a sufficiently fine approximation while avoiding unnecessary computational cost.

We also added a bandwidth sensitivity check based on cross-validation. For each player and shot outcome, shot locations were rescaled to $[0, 1] \times [0, 1]$, and Gaussian kernel density estimators were fitted over a 20-point logarithmic grid of candidate bandwidths from 10^{-3} to 1. Each candidate bandwidth was evaluated using five-fold cross-validation, with the selected bandwidth maximizing the average held-out log predictive density. The resulting densities showed spatial patterns similar to those obtained with Silverman's rule, indicating that the main features of the analysis are not driven by the particular rule-of-thumb bandwidth used in the primary results.

We have therefore clarified that the grid and bandwidth are numerical implementation choices in the functional-density representation. Unlike fixed spatial bins or manually defined shot zones, they do not determine the substantive categories being analyzed, and the added sensitivity checks show that the main interpretations are stable to reasonable alternatives.

COMMENT # 0.4

Related to the previous point, it is clear that densities are able to examine patterns without the influence of volume. However, how would removing volume information prove valuable in making performance-related decisions in terms of coaching, scouting, or training, for example? Are there other solutions that could be more effective, such as stratifying by volume rather than removing it altogether? Please address this more completely in the discussion.

Reply:

Ed To Do: Double check and either reference discussion or add part. Our density-based representation is intended to characterize where a player tends to shoot, conditional on the player having taken shots, rather than how often the player shot shoots overall. This is useful in making performance-related decisions when the goal is to compare spatial shooting tendencies independently of playing time, number of games or usage. For example, two players with very different total shot counts can still have similar spatial tendencies. Our approach allows this to be detected.

We should add something in the discussion.

We could add intensity surfaces as a third coordinate in the vector-valued stochastic process. I am not sure it is a good idea to stratify by volume, because the estimation of the densities depend on the number of shots we have for the player. So, players with less shots will have their densities less accurately estimated. Also, how do we choose the strate? Alternatively, shot volume could also be included as an additional feature in the clustering step.

COMMENT # 0.5

The choice of $k = 5$ *is clearly not data-driven. What would a true data-driven clustering look like? Certainly there are more than 5 shot pattern styles, so what information can we really gather from the clustering beyond observing that they clusters with* $k = 5$ *are poorly aligned with player position? Also, although several metrics on the clustering are reported (ARI, Silhouette score), the relevance of the values is not discussed or their bearing on the reliability and impact of the clustering results.*

Reply:

We agree that the choice of $k = 5$ is not data-driven. Its purpose is to provide a controlled comparison with the five conventional NBA position groups, not to claim that there are exactly five natural shooting styles. We have clarified this point in the revised manuscript and added a new *Data-driven clustering* subsection.

In that subsection, a fully data-driven version of the clustering keeps the same MFPCA score representation, the same k-medoids algorithm and the same distance matrices, but selects the number of clusters by maximizing the average Silhouette coefficient over $k = 2, \ldots, 10$. With equal weights for the four MFPCA scores, the maximum Silhouette coefficient is 0.18 and is obtained for $k = 3$. The three clusters contain 41, 57 and 75 players, with Norman Powell, Coby White and Dennis Schröder as medoids. With variance-based weights, the maximum Silhouette coefficient is 0.54 and is obtained for $k = 2$. The two clusters contain 99 and 74 players, with Julius Randle and Jordan Poole as medoids. Because the first component explains most of the variation, the variance-weighted data-driven clustering is mainly a split along the first score.

The negative Silhouette coefficients for the NBA position labels (-0.05 and -0.1, depending on the distance matrix) reinforce that conventional positions are not compact or well-separated groups when evaluated using shooting-pattern distances. For the five-cluster k-medoids solutions, the equal-weighted Silhouette coefficient of 0.04 indicates weak separation and supports interpreting those clusters as descriptive summaries rather than as reliable evidence of five isolated shooting types. The variance-weighted Silhouette coefficient of 0.4 indicates a clearer partition, but one driven mainly by the dominant components. Thus, the $k = 5$ analysis is useful for comparison with NBA positions, while the data-driven analysis suggests that the strongest intrinsic grouping in the MFPCA score space is coarser.

Reviewer # 1

We thank the reviewer for their thorough and detailed comments. Below is our point-by-point reply.

Comment # 1.1

Abstract

- *Overclaims novelty; FDA-based spatial modeling exists and differentiation from prior intensity-surface approaches is insufficiently clarified.*
- *No explicit statement of key methodological limitations (e.g., normalization removing volume/efficiency information).*
- *Practical contribution vs methodological novelty not clearly separated.*

Reply:

- The paper does not claim to invent FDA, spatial smoothing or basketball shot-chart modelling in general. Rather, it presents a novel application of the FDA approaches in an NBA data analysis setting (i.e., the representation of NBA shot charts as bivariate functional data on a two-dimensional domain). The Introduction section explicitly positions the work within the context of other approaches to statistical modelling in applications of this type.
- We address the limitations comprehensively in the Discussion section (with some thought on how they can be addressed going forward). We believe that it is reasonable for such limitations to appear in the Discussion section rather than in the Abstract.
- I don't understand the comment. **Ed**: I can add a comment to main text abstract to satisfy this

Comment # 1.2

Introduction and Background

- *Motivation for density representation vs intensity-based models is weakly justified; discretization critique is not convincing since FDA also requires discretization.*
- *Limited positioning within modern representation learning or embedding-based player typologies.*

- *Missing discussion of role-dependent contextual factors (pace, lineup context, possessions, play types).*
- *Additional reference that could be added to the manuscript is: https://www.mdpi.com/2504-4990/6/3/102 and https://www.mdpi.com/2078-2489/16/8/699*

Reply:

- Our point is not that numerical evaluation on a grid disappears. Actually, any practical implementation of a continuous model requires numerical evaluation. The distinction is instead about the object that is being modeled. Yes, this is the key FDA thing – I am happy to revise this comment.
- We propose a paper focused on functional representations of shot charts, not a survey of all modern embedding methods for player representation. The Introduction section situates the work in the relevant basketball literatures.
- We say explicitly that the paper focuses on shooting behaviour. The absence of role-dependent contextual factors is stated in the Discussion section.
- We don't see the connection between the proposed articles and ours, except they are talking about basketball. Also, we won't cite predatory journals.

COMMENT # 1.3

Methodology

- *Kernel density estimation choices (Gaussian kernel, Silverman bandwidth) insufficiently validated or sensitivity-tested.*
- *Grid resolution arbitrary without robustness analysis.*
- *Normalization removes shot volume and efficiency, limiting interpretability of components.*
- *MFPCA assumptions may violate density constraints (non-negativity, unit integral) — acknowledged but not resolved.*
- *Choice of* $k = 5$ *clusters driven by court positions rather than data-driven criteria.*
- *Limited justification for* k*-medoids vs alternative clustering or manifold methods.*

Reply:

- A Gaussian kernel with Silverman bandwith is a standard default for smooth density estimation. As paper's primary methodological contribution is not a new KDE procedure, we consider the density estimation step as a reasonable preprocessing device.

- Our grid is adjustable, which is the appropriate way to present an implementation parameter. Ed to add.

- Our goal is to compare spatial shot patterns across players, not to conflate style with playing time, shot volume, ... We state this explicitly in the paper and that we prefer the normalized representation for fair comparison. Volume and efficiency are important, but they are different objects.

- We agree with the mathematical point, but we disagree that it undermines the results of the paper. Principal components are deviations from the mean and make take negative values. We explicitly acknowledges the density-constraint issue by citing relevant work, and note that reconstructions can be projected back into density space if required.

- OK, what's your point?

- We do not agree that the choice is under-justified. We gave a practical reason: the k-medoids algorithm returns actual players as cluster representatives, which improves interpretability here. This is aligned with the emphasis of the paper on interpretable player typologies.

COMMENT # 1.4

Results

- *Interpretation largely descriptive; lacks statistical validation or hypothesis testing.*

- *Low cluster agreement (ARI = 0.04–0.05) not deeply analyzed — could indicate model inadequacy rather than novel insight.*

- *Silhouette scores suggest weak cluster structure but implications are under-discussed.*

- *No external validation (e.g., predictive performance, downstream task).*

Reply:

- The paper is exploratory. The goal of MFPCA is to produce interpretable dominant modes of spatial variation, not to test a null hypothesis. The results are not merely narrative: we report the variance explained by each component, ARI values, Silhouette coefficients, cluster compositions and representative medoids.

- I don't know what to say. **Ed**I am happy to take this comment and expand on it.

- The equal-weight solution has a low silhouette score, but the variance-weighted solution has a silhouette of 0.4, which is better than the NBA labellings using the same distance matrices, which are negative. The results show that the apparent structure depends on the weighting scheme, which is itself informative. The manuscript already reports those values.

- This is not a predictive paper. We are not concerned by that. Edd to add short point on how we might validate clusters.

COMMENT # 1.5

Discussion

- *Claims of interpretability and practical relevance not empirically demonstrated (e.g., coaching decisions, scouting outcomes).*

- *Limited comparison against baseline models or alternative representations.*

- *Absence of multimodal or contextual variables despite acknowledging importance.*

Reply:

- Interpretability is demonstrated in the way the method is constructed and presented. We do not claim validated front-office performance gains. We claim potential uses, which is appropriate here. We don't really have the resources or scope to do this... we don't work in a basketball club.

- We reviewed the main other approaches and explained the differences with ours. We consider that conceptual differentiation with empirical illustration is adequate here.

- The aim of the paper is to build a typology of players based on their shooting behaviour. While it would be interesting to add other variables, it will require to develop another method.

COMMENT # 1.6

Conclusion and future work

- *Future work suggestions are broad but not technically grounded (e.g., multimodal integration lacks methodological direction).*
- *Does not address scalability, computational complexity, or real-time applicability.*
- *Key methodological limitation (density normalization) insufficiently emphasized.*

Reply:

Ed: Again, I can make these a little more gentle, but to be honest this reviewer is wrong here.

- We do not agree. It is not the purpose of the paper to give methodological direction on how to integrate multi-model data.
- We do not believe real-time applicability is required here. Moreover, we make no such claims.
- This limitation is acknowledged in the Results section and in the last paragraph of the Discussion section. We also discuss how it can be addressed going forward.

COMMENT # 1.7

Technical Writing

- *Some redundancy in explaining FPCA concepts.*
- *Overly descriptive narrative in results; could benefit from more concise statistical framing.*
- *Terminology occasionally inconsistent between "shooting frequency," "density," and "intensity."*
- *Figures interpretation sometimes qualitative rather than quantitative.*
- *Keywords are not precise and should be in alphabetical order.*

Reply:

- Some repetition are intentional because the paper sits at the intersection of functional data analysis and sports analytics. Not all readers will be equally comfortable with FDA terminology, so moderate repetition improves accessibility.

- Spatial principal components are inherently geometric objects, and the dominant modes of variation can be read and interpreted on the court.

- The word "intensity" is only used in the Introduction and refer to the intensity surfaces developed in other papers. The term "shooting frequency" appears three times to describe the first components.

- The figures are intended to make the functional components and cluster medoids interpretable. Interpretable spatial representation are a core of FDA representation.

- The revised keywords, now listed in alphabetical order, are: *Basketball analytics*, *Clustering*, *Functional data analysis*, *Shot charts* and *Shot-density estimation*.

Reviewer # 2

Comment # 2.1

> *Alternative clustering approaches I think the authors should include the discussion of alternative clustering methods in the main text (or in the appendix).*

Reply:

We thank the reviewer for the suggestion. In the revised manuscript, we have added a discussion of alternative clustering approaches. We also clarified why the k-medoids algorithm was selected for the main analysis. Specifically, the k-medoids algorithm has the advantage that each cluster center is an actual player rather than an artificial average point. This makes the resulting clusters easier to interpret in this context, because each cluster can be summarized by a representative player.

We have also clarified that the choice of the method is not intrinsic to the proposed framework. The representation provides a general low-dimensional summary of each player's shooting pattern and different clustering algorithms (or other statistical methods) could be applied to the representation depending on the goal of the analysis. For example, hierarchical clustering is useful when we want to study nested similarities between players; mixture models are useful when a probabilistic assignment is desired and a density-based method is useful to identify outliers.

We have added the following sentences to the Discussion section:

> Although, we use the k-medoids clustering algorithm in our analysis, the MFPCA representation is not tied to a particular clustering algorithm. Actually, once each player's shot chart has been summarized by a finite vector of scores, all of the standard clustering algorithms could be applied. For example, k-means clustering could be used to identify groups around average score profiles, hierarchical clustering could show nested similarities among players, mixture models could provide probabilistic cluster memberships and density-based methods could be use in case of outliers. We use the k-medoids algorithm because the center of each cluster is an actual player from the NBA, rather than an artificial mean profile. This provides a direct basketball interpretation of each cluster through its medoid.

Comment # 2.2

> *Since the data span multiple seasons (2018-19 through 2023-24), shot selection can change across seasons (and also within a season), due to numerous reasons (e.g., changes in team,*

coaching, usage rate, playing style, etc.). This is perhaps outside the scope of the paper, but I'm interested in seeing a discussion on whether your method can be extended to handle this. I think having a couple of paragraphs on this in the Discussion would make the paper more intriguing to interested readers. For example, young players may enter the league without a reliable 3-point shot and therefore prefer scoring near the basket. Over time, they may develop a consistent 3-point shot and become more comfortable taking 3s. A similar pattern may apply to centers. Players such as Al Horford and Brook Lopez once favored post-ups and shots around the rim but later expanded their range and became good at 3-point shooting. (These are just examples of the top of my head; these developments by Horford and Lopez occurred before the sample of seasons considered in this paper.) Alternatively, have you also considered clustering at the player-season level instead of just one single aggregated shot chart for each player? Extending this to a player-season level could further validate your finding of empirical clusters having low agreement with official positions. For instance, it could help quantify the role/positional evolution of a player (related to what I described above).

Reply:

We thank the reviewer for this suggestion. We agree that aggregating shots across multiple seasons, while providing a stable summary, may hide potentially meaningful temporal changes. A similar procedure could be applied after redefining each observational unit from "player" to "player-season". The resulting MFPCA scores could then be analyzed longitudinally, considering that the observations are not independant. We could then model score trajectories as a function of age or something else. Clustering the player-season scores would also allow players to move between shooting-style groups over time and quantifying role evolution. While being outside the scope of the paper, we added a paragraph in the Discussion section explaining how the proposed method could be extended to account for season-to-season variability.

We have added the following sentences to the Discussion section:

> We aggregated shooting behavior across the 2018-2019 through 2023-2024 regular season. However, this aggregation may mask meaningful season-to-season changes in shooting behavior, as a player's shot selection can evolve with time due to development, injuries, coaching strategy, etc. A natural extension of our framework would be to perform the analysis at the player-season level rather than at the aggregated player level. In this setting, the MFPCA scores could be used to represents the evolution of shooting-style over time. This could help to quantify the temporal evolution of a player's role, identify players whose shooting patterns change across seasons, and examine whether these changes correspond to team

changes, coaching changes or something else. This could also strengthen the comparison between empirical shooting-pattern clusters and conventional NBA positions. We could assign a cluster to each player-season and study the stability of these assignments.

Great reply but just TBC on last point. . . think we have it wrong way around (allocate player-season to cluster rather than cluster to player-seasondots)

REFERENCES

Ramsay, J. O., & Silverman, B. W. (2005). *Functional Data Analysis*. Springer.